S2-1

Towards improved species identification in soundscapes

Mario Lasseck

Affiliation of presenting author:

Technische Universität Chemnitz, Germany

Automatic species identification in natural soundscapes remains a challenging task, despite recent advances in deep learning and the growing availability of bioacoustic training data. While large-scale datasets have helped improve classifier performance, current systems still often struggle to reliably detect certain species across diverse and acoustically complex environments.

S2-2

Scaling vision transformers for acoustic biodiversity assessments

Isabella Bermón Rojas, Claudia V. Isaza, Laura E. Molles, and Carol L. Bedoya

Affiliation of presenting author:

SISTEMIC research group, Faculty of Engineering, University of Antioquia-UdeA, Colombia

Methods for rapid acoustic biodiversity assessments such as acoustic indices, deep embeddings, and unsupervised learning algorithms offer a way of capturing complex ecological patterns while reducing effort of analysis. However, these methods have significant limitations, including their inability to isolate species-specific information from the acoustic community and difficulty interpreting outcomes. Vision Transformers offer a promising pathway to solve this problem, as these can generate accurate predictions for a large number of classes. Nevertheless, the extensive number of species requires scaling up model capacity to improve performance and generalization capabilities, facing computational demands and training data limitations. We evaluate the scalability of vision transformers for all-encompassing acoustic species recognition, that is, rapid and exhaustive biodiversity assessments. We propose a new time-frequency representation to integrate acoustic data from different taxonomic groups into a single model and test the effects of model size, data size, pre-training, and compute on species detection accuracy. We scale up vision transformers to be deeper and wider, enhancing their capacity to learn bioacoustic features. The evaluation uses an extensive dataset of 265,000 recordings covering 456 species from New Zealand. This dataset includes a comprehensive range of native and invasive fauna, including 99.7% of birds, 100% of frogs, 84.2% of mammals, 13.9% of cicadas, and 11.48% of crickets and grasshoppers. Scaling up vision transformers significantly reduces the need for species-specific models and improves the efficiency of biodiversity monitoring. We also discuss the applicability of deep supervised bioacoustic models and question their significance in conservation and bioacoustic research.

S2-3

Pied flycatcher males and AI-based models rely on similar features for song discrimination at the population level

Mario Gallego-Abenza, Quanxiao Liu & David Wheatcroft

Affiliation of presenting author:

Department of Zoology, Stockholm University, Sweden

Bird song, an important mating trait in birds, has been extensively studied in animal communication research. Over the last four decades, significant research has focused on the emergence and consequences of population differences in song and how these are perceived by individuals in natural settings. In oscine birds, song learning is suggested to speed up the process of song divergence, where distinct component sounds, i.e., syllables, are shown to differ across populations. Whereas traditional acoustic analyses have demonstrated overall spectral dissimilarities across populations, novel cutting-edge AI tools can now provide finer-scale information on key features that differentiate populations. In this study, we used an AI-based approach to identify such features across six populations of pied flycatcher (Ficedula hypoleuca) previously shown to have acoustic dissimilarity. Custom explainable AI models suggest the first part of the song, i.e., first syllables, as key features when classifying songs to the population level. To investigate whether there is an alignment between AI models and the birds´ own perception of songs, we played back songs to males manipulated to contain either local or foreign syllables at the beginning and/or end of the songs. We found that males responded stronger to songs whose first syllables were local, regardless of whether the second part was local or foreign. Our results, confirming AI results in the field, enhance the implication of novel techniques to generate biologically relevant hypotheses.

S2-4

Using AI to Crack the Code of Hawaiian Monk Seal Chatter

Jeppe Have Rasmussen, Jillian M. Sills, Kirby Parnell, Colleen Reichmuth

Affiliation of presenting author:

University of Copenhagen, Denmark and University of California, Santa Cruz, USA

The Hawaiian monk seal is a critically endangered species with only about 1600 extant individuals. Earlier studies show that mature males produce at least 6 types of underwater calls, usually in bouts of two or more, with sound production linked to reproductive status. Studying monk seal vocal behavior can: 1) improve understanding of the structure and function of their low-frequency vocalizations and 2) enable development of an automated call detector, thereby aiding in conservation efforts by using vocalizations as a proxy for the presence of free-ranging seals. Using a manually annotated dataset of 4989 discrete calls obtained in zoological settings, we trained a YOLO deep-learning network that obtained an accuracy of 95 % in detecting and classifying calls into the previously defined types. The detector processed 4 years of recordings and retrieved 49,372 calls from two adult male seals. This extensive, verified dataset—unique for marine mammals—allowed us to investigate vocal patterns in activity in relation to physiological status as well as call sequencing within bouts. Not only did the detector perform well on 10 % unseen data, but all trends in the manually annotated data were consistent with those observed in this large AI-extracted data set. For the detection of animals in the wild we combined our high-quality calls with recordings from the native ecosystem of the seals to create representative sound clips, thereby supporting acoustic based conservation for this vulnerable species.

S2-5

But what does the dog say? Toward a modular machine learning pipeline for canine bioacoustics research

Lászlo Róbert Zsiros, Nour Eddin Mohab, Franceska Furik, Tünde Aida Larionov, Petra Szögi, Tamás Faragó

Affiliation of presenting author:

BARKS Lab, Department of Ethology, Eötvös Loránd University, Hungary

Birds have BirdNET. Bats have SonoBat. Rodents have DeepSqueak. But dogs seem to have only limited bioacoustics machine learning support — let alone large, research-grade annotated audio datasets. This gap is especially striking given that canine vocalizations are ubiquitous in the human soundscape. The growing scientific interest in dogs' communicative signals — both in conspecific and human-dog interactions — further underscores the need for ML-based solutions. Developing such tools comes with its own challenges, shaped both by the physical characteristics of dog vocalizations and by the variability — or lack thereof — across available recording environments. Our goal with STRELKA (Sound Training and Recognition Engine for the Lore of K9 Acoustics) is to create a modular, process-oriented pipeline to support canine bioacoustics research in both lab and field settings, and through this, contribute to building a large-scale, open-access dataset of annotated vocalizations. STRELKA is built around a flexible, HDF5-based architecture, designed to accommodate multiple modeling strategies, from supervised classifiers to transformer-based or semi-supervised models, depending on data availability and task complexity. While the current stage of development draws primarily on our Department's extensive database of recordings, the system is structured to scale, with potential integration of citizen science contributions, wearable devices, and other forms of passive acoustic monitoring. This presentation will share experiences and ongoing challenges in pipeline development, with a particular focus on early-stage classification efforts and exploratory work toward contextual and sequence-aware modeling.

S2-6

Deep learning for acoustic classification of Orthoptera and Cicadidae in Switzerland

Jaromir Kunzelmann, Dr. Tobias Peter, Dr. Matthias Nyfeler

Affiliation of presenting author:

Department of Life Sciences and Facility Management, Institute of Computational Life Sciences, Zurich University of Applied Sciences, Switzerland

Numerous species of Orthoptera and Cicadidae in Switzerland are threatened. Some are indicator species, informing about wider ecosystem health. With many of the species being vocal, passive acoustic monitoring (PAM) is a viable option for surveys. Nowadays, low-cost recording devices make it easy to collect vast amounts of audio data using PAM. In recent years, deep learning (DL) models have become increasingly powerful. Applications of DL methods to bioacoustic monitoring data have proven useful. Once trained on a sufficiently large annotated audio recording dataset, a DL model can be used to analyse unlabelled PAM recordings. Appropriate DL-based tools are needed to support the analysis of such datasets. The success of the BirdNET framework demonstrates that it is possible to offer both a DL analysis tool for technically versed users and a mobile application using DL in the background, enabling a wider audience to engage with biodiversity. We merged published Orthoptera and Cicadidae recordings with previously unpublished recordings from Switzerland. Using this large labelled audio dataset we aim to create an acoustic DL classifier for Orthoptera and Cicadidae species occurring in Switzerland. Comparing different state-of-the-art DL models and techniques, preliminary results show that it is possible to create accurate and robust acoustic classifiers for Swiss Orthoptera and Cicadidae. A future goal is to incorporate the classifier into a mobile application to support citizen science and promote environmental education. Provided that sufficient labelled data is available, this approach could be scaled to other taxa and geographic regions.